

L10 — Spring Boot REST API with JPA/ORM

10.1 Theory — Setup Steps

1. Create a Spring Boot Maven project.
2. Add dependencies:
 - spring-boot-starter-web
 - spring-boot-starter-data-jpa
 - mysql-connector-j
3. Configure MySQL in application.properties:

spring.datasource.url=jdbc:mysql://localhost:3306/studentdb

spring.datasource.username=root

spring.datasource.password=your_password

spring.jpa.hibernate.ddl-auto=update

4. Create a JPA Entity class.
5. Create a Repository using JpaRepository.
6. Create a REST Controller with GET and POST endpoints.

Embedded Tomcat: Spring Boot provides an embedded Tomcat server through the Web starter, so a separate Tomcat installation is not required.

10.2 Practical — Student Entity and Repository

Student.java

```
import jakarta.persistence.*;

@Entity

public class Student {

    @Id

    @GeneratedValue(strategy = GenerationType.IDENTITY)

    private Long id;


    private String name;

    private double cgpa;
```

```
public Student() {}

public Student(String name, double cgpa) {

    this.name = name;

    this.cgpa = cgpa;

}

public Long getId() {

    return id;

}

public String getName() {

    return name;

}

public void setName(String name) {

    this.name = name;

}

public double getCgpa() {

    return cgpa;

}

public void setCgpa(double cgpa) {

    this.cgpa = cgpa;

}

}
```

StudentRepository.java

```
import org.springframework.data.jpa.repository.JpaRepository;
```

```
public interface StudentRepository

    extends JpaRepository<Student, Long> {

}
```

10.3 Practical — REST Controller

```
import org.springframework.web.bind.annotation.*;

import java.util.List;

@RestController

@RequestMapping("/students")

public class StudentController {

    private final StudentRepository repository;

    public StudentController(StudentRepository repository) {

        this.repository = repository;

    }

    @GetMapping

    public List<Student> getStudents() {

        return repository.findAll();

    }

    @PostMapping

    public Student addStudent(@RequestBody Student student) {

        return repository.save(student);

    }

}
```

Endpoints

GET /students → Get all students

POST /students → Add a student

Result: A REST API was created using Spring Boot, JPA, and MySQL.